

GRAZZANISE, Italy

WMO: 162530

Lat: 41.061N

Lon: 14.082E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.7	0.8	-5.9	2.3	3.7	-3.9	2.7	4.7	11.5	10.9	10.3	11.0	2.4	40	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	33.3	24.5	32.1	24.2	31.0	24.0	27.8	30.6	26.7	29.8	25.7	29.0	3.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.8	30.0	25.9	21.2	29.1	24.8	19.8	28.3	88.7	31.2	83.7	30.1	79.1	29.4	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.2	6.9	-3.5	36.1	1.4	1.5	-4.6	37.2	-5.4	38.1	-6.2	38.9	-7.2	40.0		
			-4.1	29.1	1.3	1.9	-5.1	30.5	-5.8	31.6	-6.5	32.6	-7.5	34.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	8.4	8.6	10.8	13.6	17.7	22.4	24.8	25.1	21.7	17.7	13.2	9.5	(6)
(7)		DBStd	6.57	2.71	2.80	2.53	2.51	2.55	2.86	1.97	2.05	2.49	2.64	3.12	2.95	(7)
(8)		HDD10.0	196	64	55	21	2	0	0	0	0	0	9	45		(8)
(9)		HDD18.3	1480	308	273	234	142	43	4	0	4	43	154	275		(9)
(10)		CDD10.0	2450	14	15	45	112	239	372	459	469	351	239	106	29	(10)
(11)		CDD18.3	691	0	0	0	1	24	126	201	211	105	23	2	0	(11)
(12)		CDH23.3	6353	0	0	0	15	204	1153	2017	2097	748	116	3	0	(12)
(13)		CDH26.7	2302	0	0	0	1	35	388	796	881	192	9	0	0	(13)
(14)	Wind	WSAvg	2.6	2.8	2.9	3.0	2.6	2.6	2.5	2.4	2.3	2.4	2.5	2.8	2.9	(14)
(15)	Precipitation	PrecAvg	888	91	72	84	82	47	31	23	27	78	102	136	109	(15)
(16)		PrecMax	1301	259	157	267	167	114	89	63	136	167	247	341	196	(16)
(17)		PrecMin	679	12	13	0	29	6	3	1	2	18	9	55	7	(17)
(18)		PrecStd	169	59	40	56	30	26	26	16	31	39	63	62	49	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	17.8	21.1	25.2	30.0	34.1	34.8	35.8	32.1	27.9	23.2	18.8	(19)
(20)		MCWB	14.5	13.6	14.9	18.1	21.1	22.9	25.1	24.6	22.2	20.2	18.8	16.0		(20)
(21)		2%	DB	15.6	16.0	18.9	22.8	27.1	32.0	33.1	33.8	30.1	25.9	21.2	16.9	(21)
(22)		MCWB	13.1	12.6	14.6	17.4	19.9	23.1	24.8	24.9	22.9	19.9	17.5	14.2		(22)
(23)		5%	DB	14.8	15.0	17.1	20.9	25.2	30.2	32.0	32.2	28.9	24.8	19.9	15.5	(23)
(24)		MCWB	12.3	12.2	13.7	16.1	19.6	22.8	24.3	24.8	22.5	19.6	16.6	13.2		(24)
(25)		10%	DB	13.2	14.0	15.9	19.2	24.0	29.0	30.8	31.0	27.2	23.1	18.2	14.8	(25)
(26)		MCWB	11.0	11.2	12.9	15.8	19.0	22.6	24.1	24.7	21.7	18.8	15.4	12.5		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	15.2	17.2	20.0	23.8	26.9	29.5	29.9	27.1	23.6	20.1	16.5	(27)
(28)		MADB	16.7	17.0	19.3	23.0	26.9	29.8	32.6	32.6	29.3	25.6	22.1	18.2		(28)
(29)		2%	WB	13.7	13.8	15.4	18.3	22.0	25.7	27.9	28.0	25.4	21.9	18.4	14.8	(29)
(30)		MADB	15.3	15.3	17.5	21.2	25.1	29.1	30.7	30.9	28.1	24.3	20.4	16.2		(30)
(31)		5%	WB	12.7	12.7	14.3	17.3	20.6	24.7	26.7	26.8	24.0	20.7	17.1	13.8	(31)
(32)		MADB	14.2	14.4	16.4	20.1	23.8	28.3	29.8	30.1	26.9	23.1	19.3	15.3		(32)
(33)		10%	WB	11.5	11.6	13.4	16.1	19.5	23.7	25.6	25.8	22.7	19.5	15.8	12.7	(33)
(34)		MADB	13.1	13.4	15.3	18.7	22.8	27.3	28.8	29.2	25.8	22.1	17.9	14.3		(34)
(35)	Mean Daily Temperature Range	MDBR	8.7	9.4	9.7	10.5	11.2	11.2	11.2	11.7	10.8	10.2	8.9	8.7	(35)	
(36)		5% DB	MADB	9.5	10.3	11.6	12.8	13.4	13.0	13.3	12.2	11.7	10.2	9.4	(36)	
(37)		MCWBR	7.5	7.8	8.5	8.9	8.2	7.3	7.6	8.0	7.5	7.7	7.5	7.3	(37)	
(38)		5% WB	MADB	8.5	9.0	10.4	11.7	12.1	11.7	11.2	12.0	11.0	10.3	9.5	8.2	(38)
(39)		MCWBR	7.4	7.6	8.3	9.1	8.7	8.0	8.4	8.6	8.1	8.1	7.8	7.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.372	0.412	0.456	0.454	0.450	0.447	0.447	0.436	0.403	0.380	0.350	(40)	
(41)		taud	2.436	2.344	2.230	2.118	2.180	2.240	2.259	2.266	2.283	2.360	2.355	2.427	(41)	
(42)		Ebn at Noon	799	828	831	818	829	831	829	818	799	784	745	760	(42)	
(43)		Edh at Noon	80	101	126	151	146	137	134	129	119	99	87	76	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.80	2.60	3.84	5.11	6.25	7.15	7.31	6.44	4.70	3.27	2.06	1.63	(44)	
(45)		RadStd	0.18	0.28	0.45	0.38	0.45	0.33	0.39	0.40	0.22	0.30	0.20	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.73	N/A	N/A	+1.07	+1.70	+2.12	N/A	-172	+306	+145
(47)	Regional (18 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-79	+131	+68

Nomenclature: See separate page